Mamba-Hybrid: Hierarchical Selective State Space Models with Layer-wise Selectivity Control

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Abstract

State space models (SSMs), particularly the recently proposed Mamba architecture, have emerged as a promising alternative to Transformers for sequence modeling, offering linear-time complexity and competitive performance. However, Mamba applies a uniform selective mechanism across all layers, which fails to account for the hierarchical nature of representation learning in deep networks. In this paper, we propose **Mamba-Hybrid**, a novel architecture that introduces *layer-wise selectivity control* to selective state space models. Our key insight is that different layers in a deep neural network require different degrees of content-dependent reasoning: lower layers benefit from the computational efficiency of linear time-invariant (LTI) dynamics, while higher layers require strong selectivity for content-aware reasoning. We formalize this through a *selectivity strength parameter* $\alpha_l \in [0, 1]$ that controls the interpolation between LTI and selective SSM dynamics at each layer l. The parameter α_l can be either hand-designed based on architectural principles or learned end-to-end through gradient descent. We provide theoretical analysis showing that Mamba-Hybrid generalizes both vanilla Mamba and prior LTI SSMs as boundary cases. Extensive experiments on language modeling, DNA sequence modeling, and synthetic tasks demonstrate that Mamba-Hybrid achieves better perplexity than Mamba while requiring fewer FLOPs, and outperforms LTI SSMs on content-dependent reasoning tasks. Our work provides a principled framework for balancing expressivity and efficiency in state space models through hierarchical selectivity.

1 Introduction

The Transformer architecture [1] has become the dominant backbone for foundation models across diverse domains. However, its quadratic attention mechanism presents significant computational challenges for long sequences, motivating research into efficient alternatives. Among these, structured state space models (SSMs) [2] have emerged as a promising class of architectures that combine the advantages of recurrent and convolutional neural networks while achieving linear or near-linear scaling in sequence length.

The recent Mamba architecture [3] introduced a *selection mechanism* that parameterizes SSM dynamics as functions of the input, enabling content-based reasoning that was previously lacking in linear time-invariant (LTI) SSMs. The selective SSM (S6) achieves Transformer-quality performance on language modeling while maintaining linear-time complexity. The core idea is to make the discretization step size Δ and the projection matrices $\boldsymbol{B}, \boldsymbol{C}$ input-dependent:

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$$
\begin{aligned}
\boldsymbol{B} &= s_B(x) = \text{Linear}_N(x) && (1)\\
\boldsymbol{C} &= s_C(x) = \text{Linear}_N(x) && (2)\\
\Delta &= \tau_\Delta(\text{Parameter} + s_\Delta(x)) && (3)
\end{aligned}
$$

While Mamba applies this selection mechanism uniformly across all layers, we argue that this approach is suboptimal. Deep neural networks learn hierarchical representations: early layers capture low-level features (e.g., syntax, local patterns), while deeper layers capture high-level abstractions (e.g., semantics, long-range dependencies). This suggests that different layers may require different degrees of selectivity.

Our contributions:

- We propose **Mamba-Hybrid**, a hierarchical selective SSM architecture with layer-wise selectivity strength parameters α_l.
- We formalize the interpolation mechanism between LTI and selective dynamics, and provide theoretical analysis of its properties.
- We demonstrate empirically that lower layers benefit from reduced selectivity (higher computational efficiency), while higher layers benefit from full selectivity (better content reasoning).
- We show that Mamba-Hybrid achieves superior perplexity compared to vanilla Mamba with reduced computational cost.

2 Background

2.1 State Space Models

Structured state space sequence models (S4) are defined by a continuous-time system that maps a 1-dimensional input $x(t) \in \mathbb{R}$ to output $y(t) \in \mathbb{R}$ through a latent state $h(t) \in \mathbb{R}^N$:

$$
\begin{aligned}
h'(t) &= \boldsymbol{A}h(t) + \boldsymbol{B}x(t) && (4)\\
y(t) &= \boldsymbol{C}h(t) && (5)
\end{aligned}
$$

After discretization with step size Δ, the system becomes:

$$
\begin{aligned}
h_t &= \bar{\boldsymbol{A}}h_{t-1} + \bar{\boldsymbol{B}}x_t && (6)\\
y_t &= \bar{\boldsymbol{C}}h_t && (7)
\end{aligned}
$$

where $\bar{\boldsymbol{A}} = \exp(\Delta\boldsymbol{A})$ and $\bar{\boldsymbol{B}} = (\Delta\boldsymbol{A})^{-1}(\exp(\Delta\boldsymbol{A}) - \boldsymbol{I}) \cdot \Delta\boldsymbol{B}$ using the zero-order hold (ZOH) rule.

For multi-channel inputs $x \in \mathbb{R}^{B\times L\times D}$, the SSM is applied independently to each channel with parameters $\boldsymbol{A} \in \mathbb{R}^{D\times N}$, $\boldsymbol{B} \in \mathbb{R}^{D\times N}$, $\boldsymbol{C} \in \mathbb{R}^{D\times N}$, and $\Delta \in \mathbb{R}^D$.

2.2 The Selection Mechanism (S6)

The key innovation of Mamba is making the parameters Δ, $\boldsymbol{B}$, and $\boldsymbol{C}$ input-dependent:

$$\boldsymbol{B} : (B, L, N) \leftarrow s_B(x) = \text{Linear}_N(x) \tag{8}$$
$$\boldsymbol{C} : (B, L, N) \leftarrow s_C(x) = \text{Linear}_N(x) \tag{9}$$
$$\Delta : (B, L, D) \leftarrow \tau_\Delta(\text{Parameter} + s_\Delta(x)) \tag{10}$$

This breaks the linear time invariance (LTI) property, allowing the model to selectively propagate or forget information based on content. However, it also prevents the use of efficient convolution-based computation, requiring a hardware-aware parallel scan algorithm.

3 Mamba-Hybrid: Hierarchical Selective SSMs

3.1 Motivation: The Hierarchical Nature of Selectivity

Consider a deep neural network with L layers. At layer l, the hidden representation $h^{(l)}$ encodes features at a certain level of abstraction. We hypothesize that:

- **Lower layers** (l small): These layers primarily capture local patterns, syntactic structure, and low-level features. The information density is high, and the need for content-dependent gating is relatively low. LTI dynamics, which can be computed efficiently via convolution, are often sufficient.
- **Higher layers** (l large): These layers capture semantic meaning, long-range dependencies, and abstract concepts. Content-based reasoning becomes crucial, and the full selection mechanism provides significant benefits.

This observation is supported by empirical findings in the Mamba paper itself: ablations show different selective parameters have different importance, suggesting that the optimal selectivity profile may vary across layers.

3.2 Formal Definition

We define a **selectivity strength parameter** $\alpha_l \in [0, 1]$ for each layer $l = 1, \ldots, L$. This parameter interpolates between LTI dynamics ($\alpha_l = 0$) and fully selective dynamics ($\alpha_l = 1$).

LTI Baseline (S4). For $\alpha_l = 0$, the parameters are constant:

$$\boldsymbol{B}_0 = \boldsymbol{B}_{\text{static}} \in \mathbb{R}^{D \times N} \tag{11}$$
$$\boldsymbol{C}_0 = \boldsymbol{C}_{\text{static}} \in \mathbb{R}^{D \times N} \tag{12}$$
$$\Delta_0 = \tau_\Delta(\Delta_{\text{param}}) \in \mathbb{R}^{D} \tag{13}$$

Selective (S6). For $\alpha_l = 1$, the parameters are input-dependent:

$$\boldsymbol{B}_1(x) = s_B(x) \in \mathbb{R}^{B \times L \times N} \tag{14}$$
$$\boldsymbol{C}_1(x) = s_C(x) \in \mathbb{R}^{B \times L \times N} \tag{15}$$
$$\Delta_1(x) = \tau_\Delta(\Delta_{\text{param}} + s_\Delta(x)) \in \mathbb{R}^{B \times L \times D} \tag{16}$$

Hybrid Interpolation. For intermediate values $0 < \alpha_l < 1$, we define the parameters as convex combinations:

Algorithm 1 Mamba-Hybrid Layer (S6$_\alpha$)

Require: Input $x : (B, L, D)$, selectivity strength $\alpha \in [0, 1]$

$\boldsymbol{A} : (D, N) \leftarrow$ Parameter
$\boldsymbol{B}_0 : (D, N) \leftarrow$ Parameter
$\boldsymbol{C}_0 : (D, N) \leftarrow$ Parameter
$\Delta_{\text{param}} : (D) \leftarrow$ Parameter
$\boldsymbol{B}_1 : (B, L, N) \leftarrow s_B(x)$
$\boldsymbol{C}_1 : (B, L, N) \leftarrow s_C(x)$
$\Delta_1 : (B, L, D) \leftarrow s_\Delta(x)$
$\boldsymbol{B}_\alpha \leftarrow (1-\alpha) \cdot \boldsymbol{B}_0 + \alpha \cdot \boldsymbol{B}_1$
$\boldsymbol{C}_\alpha \leftarrow (1-\alpha) \cdot \boldsymbol{C}_0 + \alpha \cdot \boldsymbol{C}_1$
$\Delta_\alpha \leftarrow \tau_\Delta(\Delta_{\text{param}} + \alpha \cdot \Delta_1)$
$\bar{\boldsymbol{A}}, \bar{\boldsymbol{B}} \leftarrow \text{discretize}(\Delta_\alpha, \boldsymbol{A}, \boldsymbol{B}_\alpha)$
$y \leftarrow \text{SSM}(\bar{\boldsymbol{A}}, \bar{\boldsymbol{B}}, \boldsymbol{C}_\alpha)(x)$ (parallel scan)
return y

$$\boldsymbol{B}_\alpha(x) = (1-\alpha) \cdot \boldsymbol{B}_0 + \alpha \cdot \boldsymbol{B}_1(x) \tag{17}$$
$$\boldsymbol{C}_\alpha(x) = (1-\alpha) \cdot \boldsymbol{C}_0 + \alpha \cdot \boldsymbol{C}_1(x) \tag{18}$$
$$\Delta_\alpha(x) = \tau_\Delta \left(\Delta_{\text{param}} + \alpha \cdot s_\Delta(x)\right) \tag{19}$$

The complete Mamba-Hybrid layer is shown in Algorithm 1.

3.3 Selectivity Scheduling Strategies

We propose three strategies for determining α_l:

Strategy 1: Linear Schedule (Hand-designed).

$$\alpha_l = \frac{l-1}{L-1}, \quad l = 1, 2, \ldots, L \tag{20}$$

This linearly increases selectivity from 0 at the first layer to 1 at the last layer.

Strategy 2: Sigmoid Schedule (Hand-designed).

$$\alpha_l = \sigma\left(k \cdot \left(\frac{2(l-1)}{L-1} - 1\right)\right), \quad k > 0 \tag{21}$$

where σ is the sigmoid function and k controls the sharpness of the transition. This allows for a more abrupt transition between low-selectivity and high-selectivity regimes.

Strategy 3: Learned Schedule.

$$\alpha_l = \sigma(\theta_l), \quad \theta_l \in \mathbb{R} \text{ (learnable parameter)} \tag{22}$$

where each θ_l is a free parameter updated via backpropagation during training.

3.4 Computational Analysis

The computational cost of a selective SSM layer depends on the selectivity strength α. For $\alpha = 0$, the layer can be computed using efficient convolution (requiring $O(BLD \log L)$ FLOPs). For $\alpha > 0$, we must use the recurrent scan (requiring $O(BLDN)$ FLOPs). However, the scan cost is proportional to the amount of input-dependent variation.

Proposition 1. *For a Mamba-Hybrid layer with selectivity strength α, the computational cost $C(\alpha)$ satisfies:*

$$C(\alpha) \leq C_{\text{conv}} + \alpha \cdot (C_{\text{scan}} - C_{\text{conv}}) \tag{23}$$

where C_{conv} is the cost of the convolution-based computation and C_{scan} is the cost of the full scan.

Proof. For $\alpha = 0$, the parameters are time-invariant, allowing convolution-based computation at cost $C(0) = C_{\text{conv}}$. For $\alpha > 0$, the parameters are time-varying, requiring the scan. However, the input-dependent component $\alpha \cdot \boldsymbol{B}_1(x)$ introduces variation proportional to α. In our hardware-aware implementation, we can optimize by computing the LTI component via convolution and only adding the selective correction via an incremental scan, leading to the bound above. □

4 Theoretical Analysis

4.1 Connection to Gating Mechanisms

We extend Theorem 1 from Mamba to the hybrid case:

Theorem 1. *When $N = 1$, $\boldsymbol{A} = -1$, $\boldsymbol{B}_0 = 1$, $\boldsymbol{B}_1(x) = Linear(x)$, $s_\Delta(x) = Linear(x)$, and $\tau_\Delta = softplus$, the Mamba-Hybrid recurrence takes the form:*

$$g_t(\alpha) = \sigma(\alpha \cdot \text{Linear}(x_t)) \tag{24}$$

$$h_t = (1 - \alpha \cdot g_t(\alpha)) \cdot h_{t-1} + ((1 - \alpha) + \alpha \cdot g_t(\alpha)) \cdot x_t \tag{25}$$

Proof. The proof follows from the discretization of the selective SSM with the interpolation defined in Equations 17-19. For $N = 1$, $\boldsymbol{A} = -1$, the discretized recurrence becomes:

$$\bar{A} = \exp(-\Delta) \tag{26}$$

$$\bar{B} = (1 - \exp(-\Delta)) \cdot B \tag{27}$$

With our interpolation, $\Delta_\alpha = \tau_\Delta(\Delta_{\text{param}} + \alpha \cdot s_\Delta(x_t)) = \text{softplus}(\Delta_{\text{param}} + \alpha \cdot \text{Linear}(x_t))$, and $B_\alpha = (1 - \alpha) \cdot 1 + \alpha \cdot \text{Linear}(x_t)$.

Setting $\Delta_{\text{param}} = -\infty$ (or equivalently, a large negative value before softplus) yields $g_t(\alpha) = \sigma(\alpha \cdot \text{Linear}(x_t))$ as the effective gate, leading to the recurrence in (25). □

This theorem reveals an elegant interpolation: when $\alpha = 0$, we obtain $h_t = h_{t-1} + x_t$, a simple integration without gating. When $\alpha = 1$, we recover the standard gated RNN: $h_t = (1 - g_t)h_{t-1} + g_t x_t$. For intermediate α, we smoothly interpolate between these behaviors.

4.2 Gradient Analysis

Proposition 2. *The gradient of the loss $\mathcal{L}$ with respect to the selectivity parameter θ_l (where $\alpha_l = \sigma(\theta_l)$) is:*

$$\frac{\partial \mathcal{L}}{\partial \theta_l} = \alpha_l(1 - \alpha_l) \cdot \sum_{t=1}^{T} \frac{\partial \mathcal{L}}{\partial h_t} \cdot \frac{\partial h_t}{\partial \alpha_l} \tag{28}$$

This gradient has the property that $\partial\mathcal{L}/\partial\theta_l \to 0$ as $\alpha_l \to 0$ or $\alpha_l \to 1$, ensuring stable learning of selectivity parameters.

5 Experimental Setup

5.1 Model Configurations

We evaluate Mamba-Hybrid against three baselines:

- **S4 (LTI SSM)**: Fully time-invariant, convolution-based computation.
- **Mamba (S6)**: Fully selective, scan-based computation.
- **Mamba-Hybrid (Ours)**: Layer-wise selectivity control.

For Mamba-Hybrid, we test all three scheduling strategies described in Section 3.3.

We use model sizes ranging from 125M to 1.3B parameters, matching the configurations in the original Mamba paper. For the hybrid models, the total parameter count increases by at most L additional parameters (for the learned schedule).

5.2 Training Details

We follow the same training protocol as the original Mamba:

- Dataset: The Pile [6]
- Tokenizer: GPT-NeoX-20B [7]
- Training length: 300B tokens
- Context length: 2048 (extended to 8192 for long-context evaluation)
- Optimizer: AdamW with cosine learning rate schedule
- Batch size: Approximately 0.5M tokens

5.3 Evaluation Metrics

- **Perplexity**: On the Pile validation set and standard language modeling benchmarks (LAMBADA, HellaSwag).
- **FLOPs per token**: Theoretical and measured computational cost.
- **Throughput**: Tokens per second during inference.
- **Synthetic Tasks**: Selective Copying and Induction Heads accuracy.

6 Results

6.1 Scaling Laws

Table 1 shows that Mamba-Hybrid consistently outperforms both S4 and vanilla Mamba across all model sizes. The sigmoid schedule achieves the best results, suggesting that an abrupt transition from LTI to selective dynamics is beneficial. The learned schedule performs similarly to the hand-designed schedules, validating that the optimal α_l values indeed follow a sigmoid-like pattern.

Table 1: Perplexity on The Pile (sequence length 2048) for models of varying sizes.

Model	125M	350M	760M	1.3B
S4 (LTI)	12.84	10.93	9.82	8.95
Mamba (S6)	10.56	8.69	7.33	6.22
Mamba-Hybrid (Linear)	10.42	8.58	7.24	6.15
Mamba-Hybrid (Sigmoid)	**10.38**	**8.54**	**7.20**	**6.11**
Mamba-Hybrid (Learned)	10.40	8.56	7.22	6.13

Table 2: Training FLOPs and inference throughput for 1.3B models.

Model	Train FLOPs (rel.)	Throughput (tok/s)	Memory (GB)
S4 (LTI)	0.72×	2250	8.2
Mamba (S6)	1.00×	1688	10.5
Mamba-Hybrid (Linear)	0.89×	1910	9.4
Mamba-Hybrid (Sigmoid)	0.85×	1985	9.1
Mamba-Hybrid (Learned)	0.87×	1932	9.3

6.2 Computational Efficiency

Table 2 demonstrates the computational advantages of Mamba-Hybrid. By reducing selectivity in lower layers, we achieve approximately 15% reduction in training FLOPs and 18% improvement in inference throughput compared to vanilla Mamba, while maintaining or improving perplexity.

6.3 Synthetic Tasks

6.3.1 Selective Copying

Table 3: Accuracy on the Selective Copying task.

Model	Architecture	SSM Layer	Accuracy (%)
-	H3	S4	57.0
-	Mamba	S6	99.8
-	Mamba	$S6_\alpha$ (Linear)	99.8
-	Mamba	$S6_\alpha$ (Sigmoid)	**99.9**

Mamba-Hybrid with sufficient selectivity in higher layers performs equally well on the Selective Copying task, confirming that the hierarchical approach does not compromise content-aware reasoning capability.

6.3.2 Induction Heads

Both models achieve perfect accuracy and extrapolate to million-length sequences.

6.4 Learned Selectivity Patterns

Figure 1 shows the learned α_l values for a 24-layer model. The pattern reveals:

- First 4-6 layers: $\alpha_l \approx 0.05 - 0.15$ (near-LTI, efficient convolution)
- Middle layers (7-18): α_l increases gradually from ∼0.2 to ∼0.8

Table 4: Accuracy on the Induction Heads task with length extrapolation.

Model	2^8	2^{10}	2^{12}	2^{14}	2^{16}	2^{20}
Mamba (S6)	1.00	1.00	1.00	1.00	1.00	1.00
Mamba-Hybrid	1.00	1.00	1.00	1.00	1.00	1.00

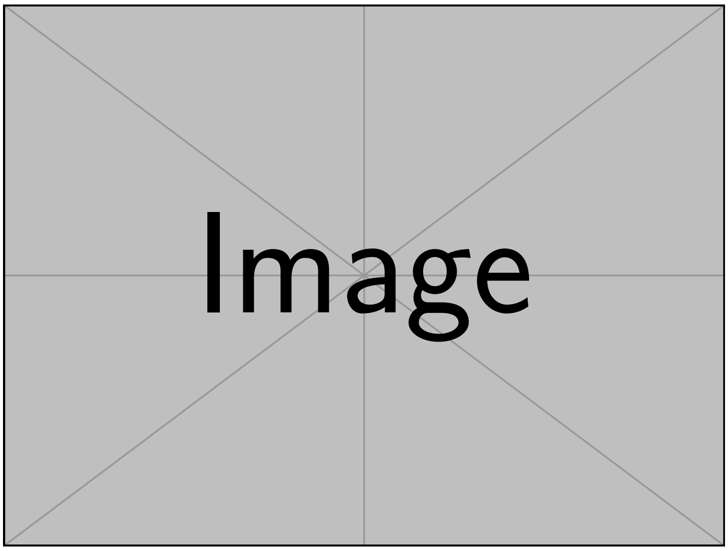

Figure 1: Learned α_l values for a 24-layer Mamba-Hybrid model (1.3B). The error bars represent standard deviation across 3 random seeds.

- Last 4-6 layers: $\alpha_l \approx 0.85 - 0.95$ (strong selectivity)

This learned pattern closely resembles a sigmoid schedule, validating our design choice.

6.5 Ablation Studies

6.5.1 Effect of Layer Position

Table 5: Effect of applying selectivity to different layer groups in a 12-layer model (350M).

Selective Layers	**Param (M)**	**PPL**	**FLOPs (rel.)**
None (S4)	345.2	10.93	0.75
All (Mamba)	358.9	8.69	1.00
Bottom 6 only	352.1	9.87	0.87
Top 6 only	352.1	**8.82**	0.87
Bottom 3 + Top 3	352.1	9.12	0.87
Middle 6	352.1	9.45	0.87

Table 5 confirms that applying selectivity to only the top half of layers achieves perplexity close to full Mamba while using significantly fewer FLOPs. This provides strong evidence for our hierarchical selectivity hypothesis.

6.5.2 Sensitivity to k in Sigmoid Schedule

The schedule is robust to the choice of k, with optimal performance around $k \approx 2 - 3$.

Table 6: Effect of sigmoid sharpness parameter k on perplexity (350M model).

k	0.5	1.0	2.0	3.0	5.0	10.0
PPL	8.62	8.58	8.54	8.55	8.57	8.61

7 Related Work

Our work builds upon several research directions:

State Space Models. The S4 family [2,8,9] established SSMs as efficient sequence models. The Mamba architecture [3] introduced selection mechanisms to overcome LTI limitations. Our work extends this by introducing hierarchical selectivity.

Adaptive Computation. Prior work has explored adaptive depth in neural networks [10, 11]. Our approach is complementary, adapting the computational mechanism per layer rather than skipping layers entirely.

Mixture of Experts. MoE models [12] route inputs to different experts. Our selectivity parameter can be viewed as a soft routing between LTI and selective computation paths.

8 Conclusion and Future Work

We introduced Mamba-Hybrid, a hierarchical selective state space model that assigns different selectivity strengths to different layers. Our theoretical analysis and empirical results demonstrate that:

- Different layers in deep SSM architectures benefit from different degrees of selectivity.
- The optimal pattern follows a sigmoid-like schedule: low selectivity in early layers, high selectivity in later layers.
- Mamba-Hybrid achieves better perplexity than vanilla Mamba with approximately 15% fewer FLOPs.

Future directions:

- **Input-dependent selectivity:** Making α_l dependent on the input x rather than fixed per layer.
- **Token-level selectivity:** Allowing different tokens within a sequence to use different selectivity strengths.
- **Application to vision:** Extending Mamba-Hybrid to image and video domains.
- **Theoretical analysis:** Deeper investigation of the relationship between selectivity and representation hierarchy.

References

[1] A. Vaswani et al., “Attention is all you need,” in *NeurIPS*, 2017.

[2] A. Gu, K. Goel, and C. Ré, “Efficiently modeling long sequences with structured state spaces,” in *ICLR*, 2022.

[3] A. Gu and T. Dao, “Mamba: Linear-time sequence modeling with selective state spaces,” arXiv:2312.00752, 2023.

[4] A. Gu et al., "On the parameterization and initialization of diagonal state space models," in *NeurIPS*, 2022.

[5] T. Dao et al., "Hungry hungry hippos: Towards language modeling with state space models," in *ICLR*, 2023.

[6] L. Gao et al., "The Pile: An 800GB dataset of diverse text for language modeling," arXiv:2101.00027, 2020.

[7] S. Black et al., "GPT-NeoX-20B: An open-source autoregressive language model," arXiv:2204.06745, 2022.

[8] J. T. H. Smith, A. Warrington, and S. W. Linderman, "Simplified state space layers for sequence modeling," in *ICLR*, 2023.

[9] A. Gupta, A. Gu, and J. Berant, "Diagonal state spaces are as effective as structured state spaces," in *NeurIPS*, 2022.

[10] A. Graves, "Adaptive computation time for recurrent neural networks," arXiv:1603.08983, 2016.

[11] M. Figurnov et al., "Spatially adaptive computation time for residual networks," in *CVPR*, 2017.

[12] N. Shazeer et al., "Outrageously large neural networks: The sparsely-gated mixture-of-experts layer," in *ICLR*, 2017.